

- 8** In a roller coaster ride, a train of mass 1200 kg and carrying a load of 400 kg is pulled up a smooth slope at a constant speed of 1.5 m s^{-1} by an electric motor to a height of 80.0 m as shown in Fig. 8.1. The train with its load is then released from rest at point A and moved to point B along the curved track.

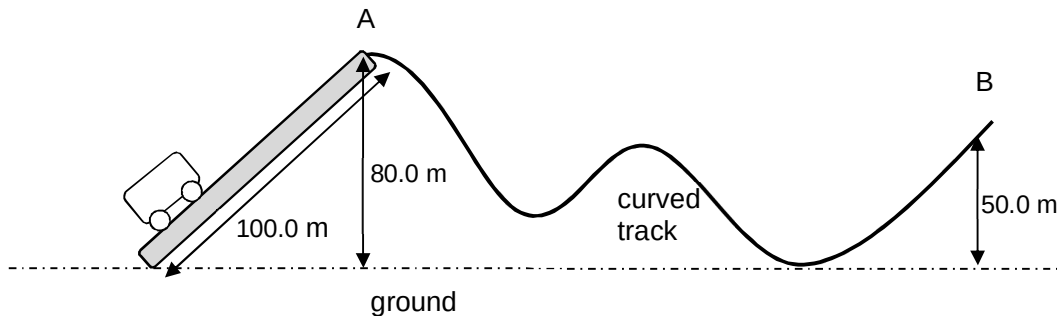


Fig. 8.1

- (a)** **(i)** Calculate the minimum force required to pull the train with its load up the slope at the given speed.

minimum force = N [2]

- (ii)** The efficiency of the motor is 80%. Calculate the electrical power that is supplied to the motor to pull the train with its load up the slope to point A.

electrical power = kW [3]

- (iii) Between A and B, the train experiences an average resistive force of 3.0 kN. If the train arrives at B with a speed of 8.0 m s^{-1} , determine the distance travelled by the train between A and B.

distance travelled = _____ m [3]

- (iv) Suggest why the resistive force on the train is unlikely to remain constant. (Ignore the resistive force due to air resistance.)

_____ [1]

- (b) In another segment of the ride, the train loop the loop in a circular track as shown in Fig.8.2. The radius of the circular motion is 25 m.

Point D is at the bottom of the circular motion.
Point E is at the right of the circular motion.
Point F is at the top of the circular motion.
Point G is at the left of the circular motion.

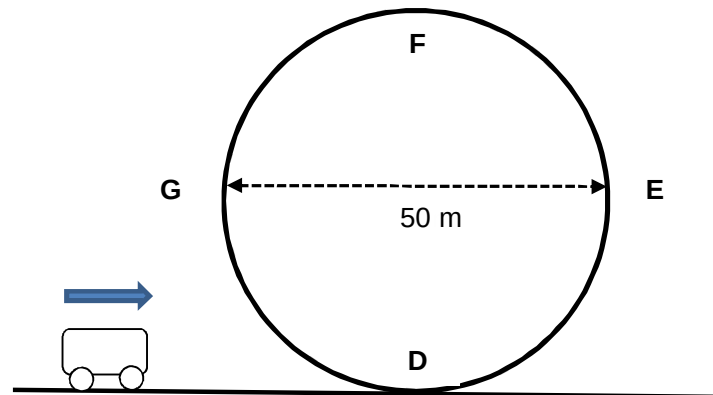


Fig. 8.2

- (i) On Fig. 8.3 below, draw the free body diagram of the train at point D and point F.

Point D



Point F



Fig. 8.3

[2]

- (ii) Determine the minimum speed of the train at point D so that it can loop the loop.

minimum speed = m s^{-1} [4]

- (c) It is suggested to keep the speed of the train constant.

- (i) On Fig. 8.4, sketch the graph to show how the normal contact force experienced by the train varies as it moves from Point D to G and back to D.

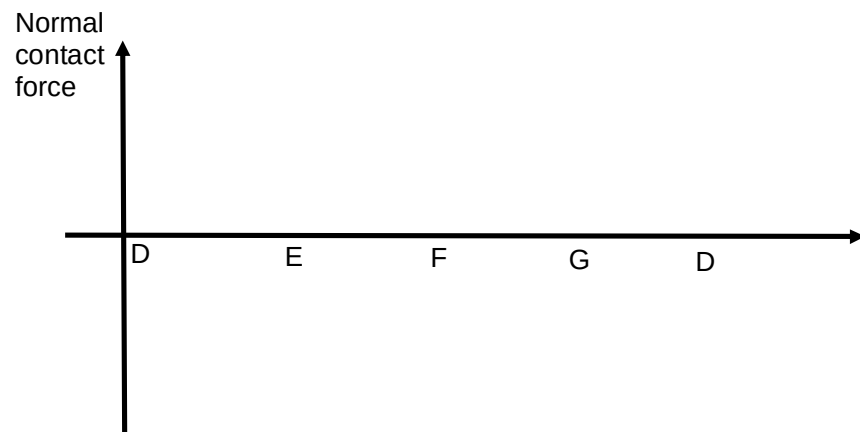


Fig. 8.4

[3]

(ii) Discuss why it is difficult to maintain the train at constant speed during the loop the loop.